

LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

CourseCode: 23CM4121

ExperimentNo: 1&2

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1. TITLE

Develop Dynamic Webpage Using JavaScript

2. AIM

To develop a dynamic webpage using JavaScript that displays a dynamic greeting, provides Dark Mode functionality, and allows users to add and delete tasks dynamically.

3. OBJECTIVES

- To understand JavaScript and its role in dynamic webpages.
- To use JavaScript with HTML elements.
- To display a greeting based on the current time.
- To implement Dark Mode using JavaScript.
- To add tasks dynamically to a list.
- To delete tasks dynamically.
- To understand DOM manipulation.
- To handle user input using JavaScript.

4. THEORY

A **dynamic webpage** is a webpage whose content or appearance can change based on user actions or other conditions.

JavaScript is used to make webpages interactive and dynamic. In this program, JavaScript performs three main tasks:

1. **Dynamic Greeting:** The current hour is obtained using `new Date().getHours()`. Based on the time, the webpage displays **Good Morning, Good Afternoon, or Good Evening**.
2. **Dark Mode:** The `classList.toggle()` method is used to add or remove the dark CSS class when the user clicks the Dark Mode button.
3. **Task Tracker:** The user enters a task, and JavaScript creates a new `<li>` element and adds it to the list. Each task also contains a Delete button.

The program uses **DOM manipulation** through methods such as `getElementById()`, `createElement()`, `appendChild()`, and `remove()`.

5. ALGORITHM / PROCEDURE

Program: Dynamic JavaScript Web Page

1. Start.
2. Create the HTML webpage with a heading, buttons, input field, and task list.
3. Get the current hour using `new Date().getHours()`.
4. Check the current hour.
5. Display Morning, Afternoon, or Evening greeting.
6. Create the `darkMode()` function.
7. Toggle the dark class when the Dark Mode button is clicked.
8. Create the `addTask()` function.
9. Read the task entered by the user.
10. Check whether the task is empty.
11. Create a new list item using `createElement()`.
12. Add the task and Delete button to the list item.
13. Add the list item to the `<ul>` using `appendChild()`.
14. Clear the input field.
15. Stop.

6. PROGRAM

JS test.js

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Dynamic JavaScript Web Page</title>
5
6      <style>
7          body {
8              font-family: Arial;
9              padding: 20px;
10             background: ■ #f4f4f4;
11         }
12
13         .dark {
14             background: □ #222;
15             color: white;
16         }
17
18         .box {
19             background: white;
20             padding: 20px;
21             max-width: 500px;
22             margin: auto;
23         }
24
25         .dark .box {
26             background: □ #333;
27         }
28
29         button {
30             padding: 8px;
31             margin: 5px;
32             cursor: pointer;
33         }
34
35         li {
36             margin: 10px;
37         }
```

```
35     {
36         margin: 10px;
37     }
38 </style>
39 </head>
40
41 <body>
42
43     <div class="box">
44
45         <h1 id="greeting">Welcome!</h1>
46
47         <button onclick="darkMode()">Dark Mode</button>
48
49         <h2>Task Tracker</h2>
50
51         <input id="task" placeholder="Enter task">
52         <button onclick="addTask()">Add Task</button>
53
54         <ul id="list"></ul>
55
56     </div>
57
58     <script>
59
60         // Dynamic greeting
61         let hour = new Date().getHours();
62
63         if (hour < 12)
64             document.getElementById("greeting").innerHTML = "Good Morning!";
65         else if (hour < 18)
66             document.getElementById("greeting").innerHTML = "Good Afternoon!";
67         else
68             document.getElementById("greeting").innerHTML = "Good Evening!";
```

```

67     else
68         document.getElementById("greeting").innerHTML = "Good Evening!";
69
70
71     // Dark mode
72     function darkMode() {
73         document.body.classList.toggle("dark");
74     }
75
76
77     // Add task
78     function addTask() {
79
80         let task = document.getElementById("task").value;
81
82         if (task == "") return;
83
84         let li = document.createElement("li");
85
86         li.innerHTML = task +
87             ' <button onclick="this.parentElement.remove()">Delete</button>';
88
89         document.getElementById("list").appendChild(li);
90
91         document.getElementById("task").value = "";
92     }
93
94 </script>
95
96 </body>
97 </html>

```

7. CODE EXPLANATION

Code	Explanation
new Date().getHours()	Gets the current hour from the system time.
if (hour < 12)	Checks whether the time is before 12 PM.
innerHTML	Changes the greeting displayed on the webpage.
function darkMode()	Defines the function for Dark Mode.
classList.toggle("dark")	Adds or removes the dark class.
getElementById("task")	Accesses the task input element.
.value	Gets the value entered by the user.
if (task == "") return;	Stops the function if no task is entered.
createElement("li")	Creates a new list item.
appendChild(li)	Adds the new task to the task list.
remove()	Deletes the selected task.
onclick	Executes a JavaScript function when a button is clicked.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open the webpage	Greeting should be displayed	Greeting displayed	Pass

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-02	Open webpage in morning	“Good Morning!” should display	“Good Morning!” displayed	Pass
TC-03	Click Dark Mode	Page should change to dark theme	Dark theme applied	Pass
TC-04	Enter Complete Assignment and click Add Task	Task should appear in list	Task displayed	Pass
TC-05	Click Delete button	Selected task should be removed	Task removed	Pass
TC-06	Click Add Task without entering a task	No empty task should be added	No empty task added	Pass

Execution Data

Input:

Task: Complete Assignment

Expected Output:

Complete Assignment Delete

9. OUTPUT

Output 1 – Dynamic Greeting

Good Evening!

Task Tracker

[Enter task] [Add Task]

Output 2 – Task Added

Task Tracker

Complete Assignment [Delete]

Output 3 – Dark Mode

The webpage background changes to dark mode when the **Dark Mode** button is clicked.

Note: Add your actual browser screenshot here showing the webpage and task list.

10. RESULT

Thus, a **dynamic webpage using JavaScript** was successfully developed. The program correctly displays a time-based greeting, provides Dark Mode functionality, and allows users to add and delete tasks

Good Evening!

Dark Mode

Task Tracker

Enter task

Add Task

- train time:7pm

Delete
- exercise-1hr

Delete
- observation and record

Delete